6061 Introduction to Deep Learning

Exercise 2 – Solutions

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This exercise sheet will not be graded and is provided solely as practice.

1. **Linear Algebra** From Mathematics for Machine Learning Chapter 2:
   (a) Exercise 2.4

   Solution:
   Online solution

2. **Derivatives** Based on the Deep Learning course from University of Tübingen.
   (a) For $a, x \in \mathbb{R}^n$, show that $\nabla_x a^T x = a^T$.

   Solution:

$$\nabla_x a^T x = \nabla_x \left(\sum_{i=1}^{n} a_i x_i \right) = \begin{bmatrix} \frac{\partial \sum_{i=1}^{n} a_i x_i}{\partial x_1} & \frac{\partial \sum_{i=1}^{n} a_i x_i}{\partial x_2} & \dots & \frac{\partial \sum_{i=1}^{n} a_i x_i}{\partial x_n} \end{bmatrix}$$
$$= \begin{bmatrix} a_1 & a_2 & \dots & a_n \end{bmatrix} = a^T$$

   (b) Let $\mathbf{f}(\mathbf{x}) = (\exp(x_1)\sin(x_3), \sqrt{x_2 + x_3}), \mathbf{f} : \mathbb{R}^3 \to \mathbb{R}^2$. Calculate the matrix of partial derivatives $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x})$.

   Solution:

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \end{bmatrix} = \begin{bmatrix} \exp(x_1)\sin(x_3) & 0 & \exp(x_1)\cos(x_3) \\ 0 & \frac{1}{2\sqrt{x_2+x_3}} & \frac{1}{2\sqrt{x_2+x_3}} \end{bmatrix}$$

   (c) Let $a : \mathbb{R} \to \mathbb{R}$ and define $\mathbf{f}(\mathbf{x}) = (a(x_1), \dots, a(x_n)), \mathbf{f} : \mathbb{R}^n \to \mathbb{R}^n$. Calculate the matrix of partial derivatives $\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x})$. What property does the resulting matrix have?

   Solution:

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}}(\mathbf{x}) = \begin{bmatrix} \frac{\partial a}{\partial x_1} & 0 & \dots & 0 \\ 0 & \frac{\partial a}{\partial x_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{\partial a}{\partial x_n} \end{bmatrix}$$

   This is a diagonal matrix.

   (d) Let $\sigma(x) = \frac{1}{1+\exp(-x)}, \sigma : \mathbb{R} \to \mathbb{R}$. Show that $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

   Solution:

$$\sigma(x) = \frac{1}{1+\exp(-x)} = \frac{\exp(x)}{1+\exp(x)}$$
$$\frac{\partial \sigma}{\partial x} = \frac{\exp(x) \cdot (1+\exp(x)) - \exp(x) \cdot \exp(x)}{(1+\exp(x))^2}$$
$$= \frac{\exp(x)}{(1+\exp(x))^2}$$
$$= \sigma(x) \cdot \frac{1}{1+\exp(x)}$$
$$= \sigma(x) \cdot (1 - \sigma(x))$$

(e) Calculate the derivative $\tanh'(x)$ with $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \tanh : \mathbb{R} \to \mathbb{R}$.

Solution:

$$\begin{aligned}\frac{\partial \tanh}{\partial x} &= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2} \\ &= \frac{(e^{2x} + 2e^x e^{-x} + e^{-2x}) - (e^{2x} - 2e^x e^{-x} + e^{-2x})}{(e^x + e^{-x})^2} \\ &= \frac{4}{(e^x + e^{-x})^2} \\ &= 1 - \tanh^2(x)\end{aligned}$$

3. **Probability**

(a) Show that $E(aX) = aE(X)$.

Solution:

$$E(aX) = \sum_x axp(x) = a\sum_x xp(x) = aE(X) \qquad \text{for discrete RV.}$$

$$E(aX) = \int axp(x)dx = a\int xp(x)dx = aE(X) \qquad \text{for continuous RV.}$$

(b) Show that $E(a) = a$.

Solution:

$$E(a) = \sum ap(x) = a\sum p(x) = a \cdot 1 = a.$$

(c) Show that $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$.

Solution:
Two facts:

$$\begin{aligned}Var(X) &= E[(X - E(X))^2] \\ &= E[X^2 - 2XE(X) + E(X)^2] \\ &= E(X^2) - 2E(X)^2 + E(X)^2 \\ &= E(X^2) - E(X)^2\end{aligned}$$

$$\begin{aligned}Cov(X, Y) &= E((X - E(X))(Y - E(Y))) \\ &= E(XY - XE(Y) - YE(X) + E(X)E(Y)) \\ &= E(XY) - 2E(X)E(Y) + E(X)E(Y) \\ &= E(XY) - E(X)E(Y)\end{aligned}$$

Use the two above facts after plugging into the definition of $Var(X + Y)$:

$$\begin{aligned}Var(X + Y) &= E((X + Y)^2) - E(X + Y)^2 \\ &= E(X^2 + 2XY + Y^2) - (E(X) + E(Y))^2 \\ &= E(X^2) - E(X)^2 + E(Y^2) - E(Y)^2 + 2(E(XY) - E(X)E(Y)) \\ &= Var(X) + Var(Y) + 2Cov(X, Y)\end{aligned}$$

(d) Is $p_X(x) = 3x^2 - e^x$ for $2 \leq x \leq 4$ a valid probability density function? If not, how could the function be modified for it to be a valid PDF?

Solution:
No, $p(4) < 0$. Probabilities have to be larger or equal to zero.
Pick an interval where the function is positive (e.g., $[1, 3]$) and then normalize.